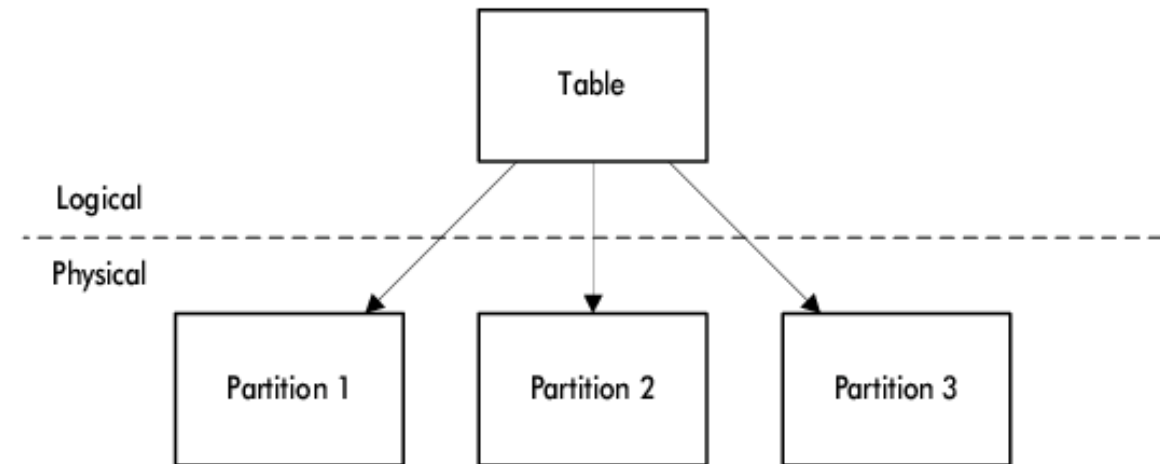


PostgreSQL Partitioning

What is Partitioning

- Split large tables into smaller pieces
- Improves performance
- Logical table interface with physical separation
- Used for scaling and performance tuning

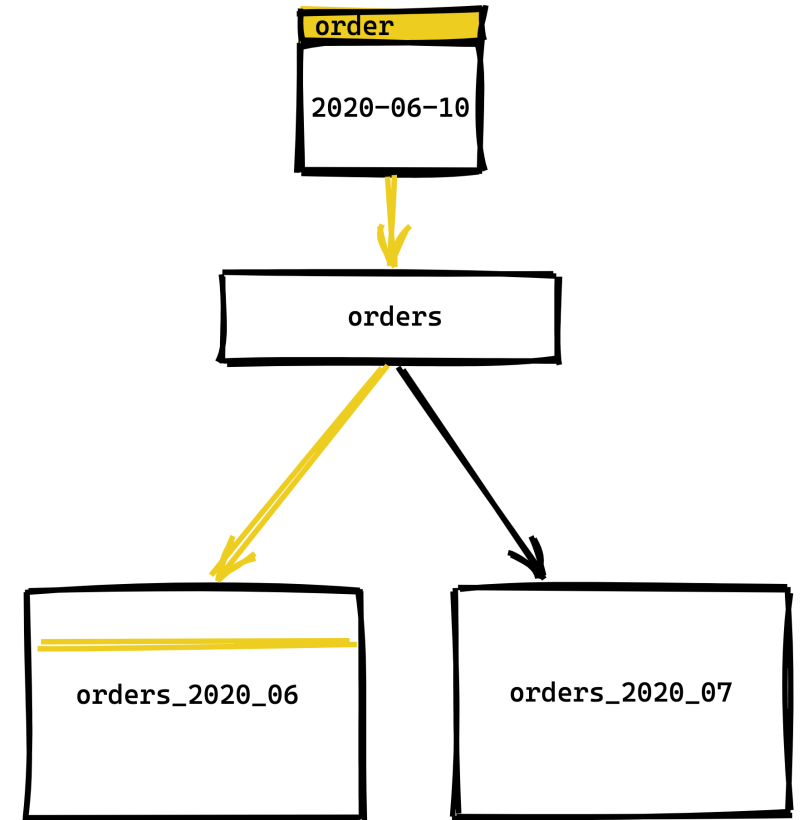


Partitioning Strategies

- Range Partitioning: Based on value ranges (e.g. dates)
- List Partitioning: Based on discrete values (e.g. types)
- Hash Partitioning: Evenly distributes rows using hash
- Multi-Level Partitioning: Combine strategies (e.g. year + type)

Key Concepts

- Parent Table: Logical interface
- Child Partition: Physical storage unit
- Partition Key: Column used for routing
- DEFAULT Partition: Catch-all for unmatched rows
- Pruning: Skipping irrelevant partitions in queries



How it works

- Insert: PostgreSQL routes row based on partition key
- Select: Planner prunes non-matching partitions
- Efficient data access when filters align with partition key

Basic Syntax

```
CREATE TABLE transactions_partitioned (  
    id INT GENERATED BY DEFAULT AS IDENTITY,  
    transaction_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    PRIMARY KEY (id, transaction_date)  
)  
PARTITION BY RANGE (transaction_date);  
  
CREATE TABLE transactions_2024_01 PARTITION OF transactions_partitioned  
FOR VALUES FROM ('2024-01-01') TO ('2024-02-01');
```

Basic Syntax

```
CREATE TABLE accounts_partitioned (  
  id      INT GENERATED BY DEFAULT AS IDENTITY,  
  account_type VARCHAR(20) NOT NULL CHECK (account_type IN ('pending', 'completed')),  
  PRIMARY KEY (id, account_type)  
) PARTITION BY LIST (account_type);
```

```
CREATE TABLE pending_accounts PARTITION OF accounts_partitioned  
  FOR VALUES IN ('pending');
```

Basic Syntax

```
CREATE TABLE companies_partitioned (  
  id    INT PRIMARY KEY,  
  ...  
) PARTITION BY HASH (id);
```

```
CREATE TABLE customers_part_0 PARTITION OF customers_partitioned  
  FOR VALUES WITH (MODULUS 4, REMAINDER 0);
```

...

```
CREATE TABLE customers_part_1 PARTITION OF customers_partitioned  
  FOR VALUES WITH (MODULUS 4, REMAINDER 3);
```


Basic Syntax

-- Detach a partition from the parent table

ALTER TABLE transactions

DETACH PARTITION transactions_2022;

-- Drop the detached table

DROP TABLE transactions_2022;

-- Attach a previously detached table (must match partition key)

ALTER TABLE transactions

ATTACH PARTITION transactions_2022

FOR VALUES FROM ('2022-01-01') TO ('2023-01-01');

Thank you

- Author: Dmytro Halchenko
- My LinkedIn: <https://www.linkedin.com/in/dmytro-halchenko-0307751b6/>
- Date: June 2025
- [Join Codeus community in Discord](#)
- [Join Codeus community in LinkedIn](#)